RSA(**R**ivest-**S**hamir-**A**dleman) Algorithm

Choose two large prime numbers, say **p** and **q**.

Calculate **n = p * q**.

Calculate **Φ(n) = (p – 1) * (q – 1).**

Choose **e**, such that

$1 < e < \Phi(n)$, and

$\gcd(e, \Phi(n)) = 1$

that is e should be co-prime with Φ(n).

$(d * e) \equiv 1 \bmod \Phi(n)$

Finally,

Public Key = (n, e) and the

Private Key = (n, d).

Encryption:

$$C = M^e \bmod n$$

Decryption:

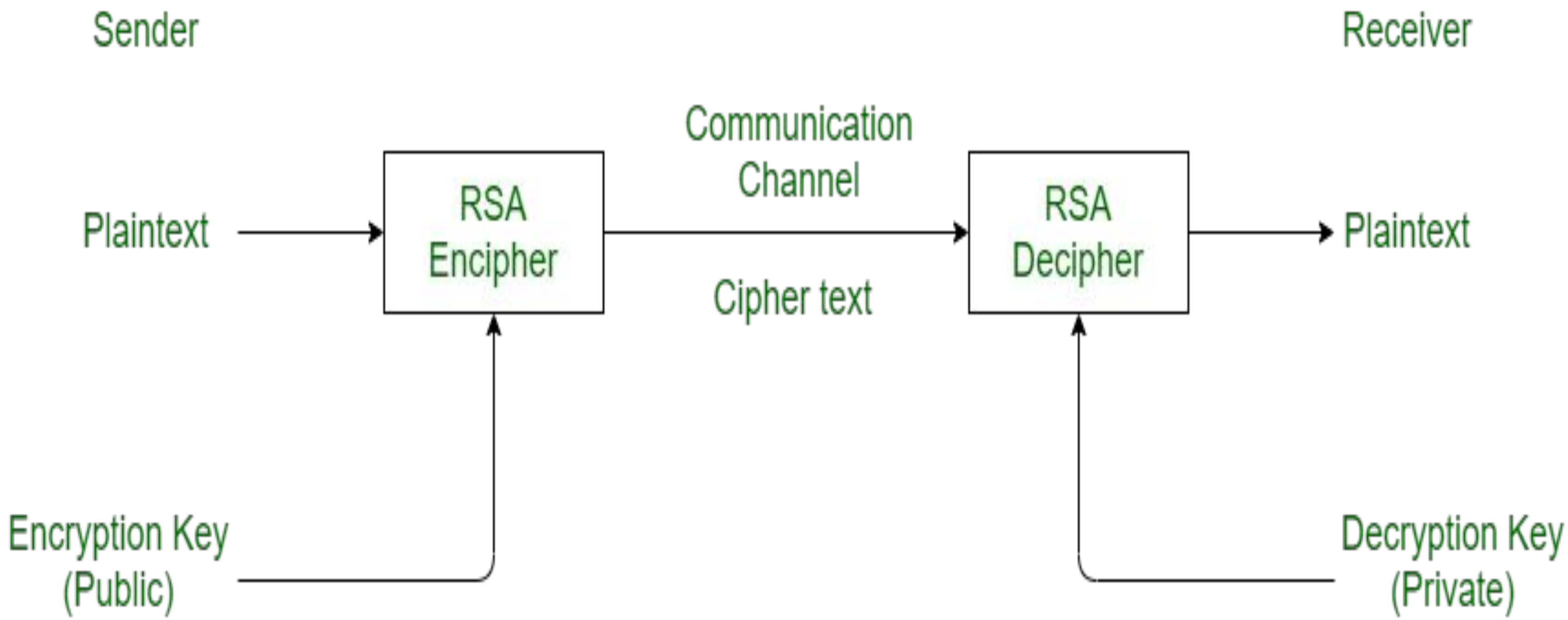
$$M = C^d \bmod n$$

What is RSA?

It is a cryptosystem used for secure data transmission.

In the RSA algorithm, the encryption key is public but the decryption key is private.

It was developed by **Rivest**, **Shamir,** and **Adleman** in 1977.


Sender
Receiver
Communication
Channel
Plaintext
RSA
Encipher
RSA
Decipher
Plaintext
Cipher text
Encryption Key
(Public)
Decryption Key
(Private)